

DHIRAJ KUMAR

Software Development Engineer | Full-Stack | DSA & System Design

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PROFESSIONAL SUMMARY

Results-driven Full-Stack Software Development Engineer with production experience building scalable MERN stack applications, RESTful APIs, and real-time WebSocket systems. Skilled in Docker containerisation, CI/CD pipelines, Google Cloud Run deployments, and API testing with Postman. Strong CS fundamentals with 300+ LeetCode problems solved, System Design knowledge, and exposure to AWS. Seeking SDE internship or full-time roles to deliver high-impact, engineering-grade software.

EDUCATION

B.Tech — Computer Science & Engineering | CGPA: 8.88 / 10.0

Sep 2023 – May 2027

Gurukul Kangri (Deemed) University, Haridwar

TECHNICAL SKILLS

Languages: Python, C++, Java, JavaScript, TypeScript

CS Fundamentals: Data Structures & Algorithms (DSA), OOP, DBMS, OS, Computer Networks, System Design

Backend / APIs: Node.js, Express.js, REST API, JWT Authentication, WebSockets, Socket.io, Postman

Frontend: React.js, TypeScript, Tailwind CSS, HTML5, CSS3, Responsive Design

Databases: MongoDB, MySQL, SQLite, Schema Design, Indexing, Query Optimization

DevOps / Cloud: Docker, Google Cloud Run, AWS (basics), CI/CD, Git, GitHub, Vercel, MongoDB Atlas

Testing / Other: Postman, Jest (basics), API Testing, Agile, Linux

INTERNSHIP EXPERIENCE

Software Engineer Intern | Yuga Yatra Retail (OPC) Pvt. Ltd.

Sep 2025 – Nov 2025

- Architected and delivered 4+ optimized REST API endpoints with query tuning and Postman-tested validation, reducing average API response time by 25% and improving SLA adherence.
- Resolved 3 critical production database failures and re-architected backend services for horizontal scaling, achieving 2x user load capacity while maintaining 100% sprint delivery rate.

Software Engineer Intern | Bireena Infotech

Apr 2026 – Present

- Engineered responsive MERN stack modules for a Hospital Management & EMR system, improving clinical workflows and data handling efficiency through RESTful API integration and Agile sprints.
- Led GitHub-based collaborative development across feature branches, ensuring seamless frontend-backend integration and deployment-ready code quality with peer-reviewed pull requests.

PROJECTS

ReLife — Divorce Support Platform — [Node.js](#) · [Express.js](#) · [React.js](#) · [MongoDB](#) · [Docker](#) · [Google Cloud Run](#) · [Twilio](#) · [Razorpay](#) [↗ Live](#)

- Built production-grade platform with JWT-based RBAC (User/Moderator/Admin), rate limiting (100 req/15 min per IP), input validation, and Helmet security headers — API tested end-to-end via Postman.
- Integrated Twilio OTP authentication and Razorpay payment gateway (UPI + QR) with server-side verification and complete order lifecycle REST APIs.
- Delivered real-time support groups across 6 categories with Chart.js admin analytics, 6-language localisation, and encrypted anonymous sessions — 98% satisfaction across 20+ users.
- Containerised with Docker, deployed on Google Cloud Run via Cloud Build CI/CD and Vercel; maintained >99% uptime with MongoDB Atlas and automated backups.

CryptoQuantix — Paper Trading Platform — [React 18](#) · [TypeScript](#) · [Vite](#) · [Node.js](#) · [Express.js](#) · [SQLite](#) · [WebSockets](#) · [JWT](#) [↗ Live](#)

- Engineered full-stack paper trading platform with real-time P&L tracking, per-second WebSocket price engine, volatility simulation, and auto-reconnection — eliminating all polling overhead.
- Implemented multi-configuration portfolio management (Conservative/Moderate/Aggressive) with independent risk parameters, asset transfer engine, and cost-basis-preserving accounting.
- Secured with JWT auth, bcrypt, Helmet HTTP headers, and CORS; integrated BTC/altcoin correlation matrix, algo backtesting, and candlestick charts via Recharts dashboards.

Smart Medical Shop — AI Health Hub — [Node.js](#) · [Express.js](#) · [MongoDB](#) · [OCR](#) · [QR Code](#) · [Voice Search](#) · [Multi-language](#) [↗ Live](#)

- Built OCR extraction pipeline using string-parsing algorithms eliminating 100% of manual prescription entry; drug interaction checker uses hash-based O(1) lookups for real-time safety checks.
- Designed REST API with QR code verification, multi-language voice search, and health reminders; sustains sub-300ms response times under concurrent load — validated via Postman test suites.

CERTIFICATIONS & ACHIEVEMENTS

Certifications: Silver Medalist NPTEL Java — IIT Kharagpur (2025) · Cloud Computing Fundamentals — IBM (2025) · Intro to Machine Learning — IIT Madras (2024) · Front-End Web Dev — IBM SkillsBuild (2024)

Competitive Programming: LeetCode 300+ problems solved (Data Structures & Algorithms) · GeeksforGeeks 200+ problems solved

Achievements: Microsoft Learn Student Ambassador · Google Cloud Arcade Facilitator · NDLI Club — IIT Kharagpur (2024) · Open Source Contributor — GitHub